

Daily Problem: 19–Feb–2026

Problem Statement

A context-free grammar is *linear* if each production contains at most one non-terminal. Show that the language of well-balanced parentheses (the Dyck language over one type of parenthesis) cannot be generated by a linear grammar.

Formally, over $\Sigma = \{ (,) \}$, let

$$D = \{ w \in \Sigma^* \mid w \text{ is a well-balanced parenthesis string} \}.$$

Prove that D is *not* a linear context-free language.

Tool: Pumping Lemma for Linear Context-Free Languages

We use the standard pumping lemma for linear CFLs:

[Pumping lemma for linear CFLs] If L is generated by a linear grammar, then there exists $p \geq 1$ such that every word $w \in L$ with $|w| \geq p$ can be written as

$$w = uvxyz$$

with

$$|vy| \geq 1, \quad |vxy| \leq p,$$

and for all $k \geq 0$,

$$u v^k x y^k z \in L.$$

(Compared to the CFL pumping lemma, the key feature is that the two pieces v and y are pumped *in sync* with the same exponent k .)

Proof that D is not linear

Assume for contradiction that D is linear. Let p be the pumping constant from Theorem .

Consider the balanced word

$$w = ({}^p) {}^p ({}^p) {}^p.$$

This word is in D and has length $4p \geq p$, so it must satisfy the linear pumping lemma. Thus we can write

$$w = uvxyz$$

with $|vy| \geq 1$, $|vxy| \leq p$, and $uv^kxy^kz \in D$ for all $k \geq 0$.

A structural observation about where vxy can lie

Write w as four blocks of length p :

$$w = ABCD \quad \text{where} \quad A = ({}^p, B =)^p, C = ({}^p, D =)^p.$$

Since $|vxy| \leq p$, the substring vxy can intersect at most two *adjacent* blocks among A, B, C, D . So there are only three possibilities:

$$vxy \subseteq A \cup B, \quad vxy \subseteq B \cup C, \quad vxy \subseteq C \cup D,$$

or it lies entirely in a single block (which is included in the above).

We will show that *in every case* there exists some k (in fact $k = 0$ will suffice) such that $uv^kxy^kz \notin D$, contradicting the lemma.

Useful property of balanced parentheses

A word $t \in \{ (,) \}^*$ is in D if and only if:

1. the total number of (equals the total number of), and
2. in every prefix of t , the number of (is at least the number of).

We will violate one of these conditions after pumping.

Case 1: vxy lies within A (only '('s) or within D (only ')'s)

If $vxy \subseteq A$, then v and y consist only of '('s, and $|vy| \geq 1$ implies we remove at least one (when pumping down to $k = 0$. Thus uxz has strictly fewer ('s than)'s overall, so it cannot be balanced. Hence $uxz \notin D$, contradicting the lemma.

Similarly, if $vxy \subseteq D$, then pumping down removes at least one), leaving more ('s than)'s overall, so again $uxz \notin D$.

Case 2: $vxy \subseteq B$ or $vxy \subseteq C$

If $vxy \subseteq B$, then v and y consist only of ')'s. Pumping down ($k = 0$) deletes at least one) from the block B , so the resulting word has more ('s than)'s overall, hence is not balanced. So $uxz \notin D$.

If $vxy \subseteq C$, then v and y consist only of '('s. Pumping down deletes at least one (, so the resulting word has more)'s than ('s overall, hence is not balanced. So $uxz \notin D$.

Case 3: vxy crosses the boundary between adjacent blocks

Now assume vxy intersects two adjacent blocks. There are three subcases.

Case 3a: $vxy \subseteq A \cup B$.

Here vxy lies within the first $2p$ symbols, i.e. within $(^p)^p$. Let r be the number of ('s removed when pumping down ($k = 0$), and let s be the number of)'s removed. Since $|vy| \geq 1$, at least one symbol is removed, so $r + s \geq 1$.

- If $r \neq s$, then the total numbers of (and) in uxz are no longer equal (because only r opens and s closes were removed), so $uxz \notin D$.
- If $r = s \geq 1$, then we removed at least one (from the initial block A and at least one) from block B (because the substring crosses the boundary and $r = s > 0$). In particular, the word uxz begins with only $p - r$ opening parentheses before encountering the remaining closing parentheses from B . Since $p - r < p$, there is a prefix of uxz that contains all $p - r$ initial opens and then $p - r + 1$ closes (coming from

the remaining part of B), so in that prefix the number of closes exceeds opens. Hence the prefix condition fails, so $uxz \notin D$.

Thus in all subcases, pumping down breaks membership in D .

Case 3b: $vxy \subseteq B \cup C$.

Here vxy lies within the middle $2p$ symbols, i.e. within $)^p($. Notice that any string in D cannot have a prefix with more $)$ than $($. But $)^p($ contains a long run of $)$'s before the next run of $($'s. Pumping down removes some substring from within this region. Again let r be the number of $($'s removed and s the number of $)$'s removed.

- If $r \neq s$, total balance fails, so $uxz \notin D$.
- If $r = s \geq 1$, then in particular at least one $)$ is deleted from the middle $)^p$ part and at least one $($ is deleted from the middle $(^p$ part. However, the prefix of the word up to the end of the remaining $)^p$ block still contains p opens from the initial $(^p$ block, but now contains $(p - s)$ closes from the middle $)^p$ block. Immediately after that, before we start reading the remaining $($'s from block C , the prefix difference (opens minus closes) equals $p - (p - s) = s \geq 1$. Now, since we also removed $r = s$ opens from block C , the remaining suffix has fewer opens available to “pay back” any later closes, and when we enter the final block $)^p$ (block D), one can check that eventually the word ends with p closes but has only p opens available after the first two blocks; deleting opens from C causes some prefix inside the final $)^p$ to have more closes than opens. Thus the prefix condition fails, so $uxz \notin D$.

(Informally: deleting matched numbers of $)$ in B and $($ in C shifts where the “matching” can happen; since the machine deletes from the central boundary, it creates a deficit before the final block of p closing parentheses.)

Case 3c: $vxy \subseteq C \cup D$.

This is symmetric to Case 3a (shifted by $2p$). The same argument shows that pumping down either breaks total balance (if different numbers of opens/closes are removed) or breaks the prefix condition within the last $(^p)$ segment. Hence $uxz \notin D$.

Final contradiction

In every possible placement of vy (since $|vy| \leq p$), pumping down with $k = 0$ produces a word $uxz \notin D$. This contradicts Theorem , which requires $uv^kxy^kz \in D$ for all $k \geq 0$.

Therefore D is not linear.

Conclusion

The language of well-balanced parentheses cannot be generated by any linear context-free grammar.